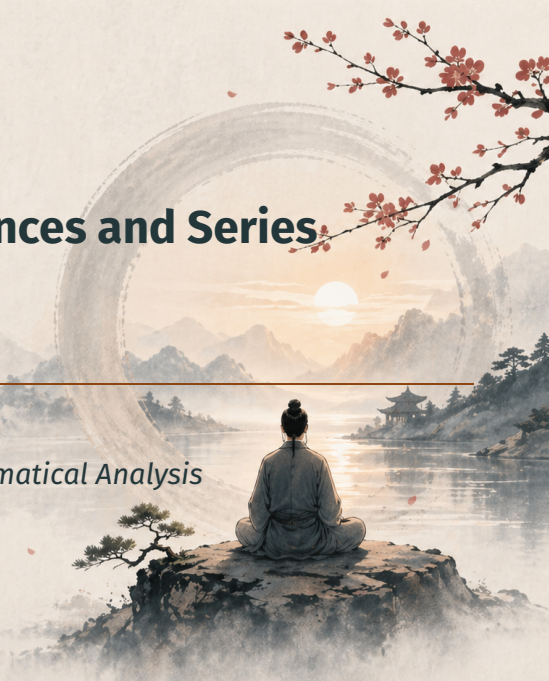


Chapter 3: Numerical Sequences and Series

Section: Upper and Lower Limits

Based on Walter Rudin's *Principles of Mathematical Analysis*



Outline

Motivation

Divergence to Infinity

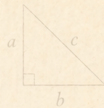
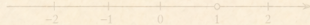
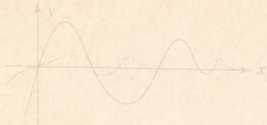
Upper and Lower Limits

Properties of the Upper Limit

Examples

Comparing Sequences

Summary



$$a^2 + b^2 = c^2$$

Motivation



$$f(x) = \sin x$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Why Upper and Lower Limits?

- Many sequences **fail to converge** (oscillation, divergence)



$$f(x) = \sin x$$



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Why Upper and Lower Limits?

- Many sequences **fail to converge** (oscillation, divergence)
- Yet they often have a clear *asymptotic ceiling and floor*



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- Many sequences **fail to converge** (oscillation, divergence)
- Yet they often have a clear *asymptotic ceiling and floor*
- **Idea:** take *sup* and *inf* of all subsequential limits



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Why Upper and Lower Limits?

- Many sequences **fail to converge** (oscillation, divergence)
- Yet they often have a clear *asymptotic ceiling and floor*
- **Idea:** take *sup* and *inf* of all subsequential limits
- **Result:** $\limsup$ and $\liminf$ **always exist** in $[-\infty, +\infty]$

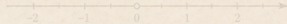


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Divergence to Infinity



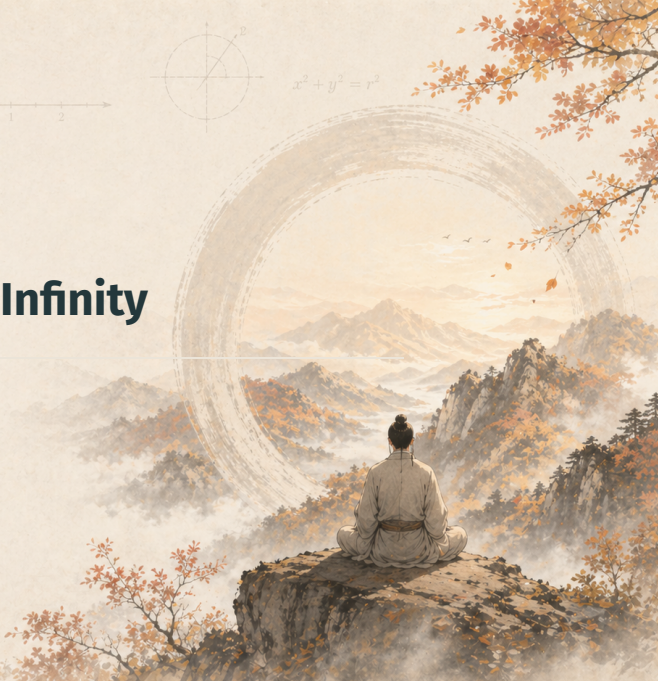
$$x^2 + y^2 = r^2$$



$$a^2 + b^2 = c^2$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$



Definition 3.15: Divergence to $\pm\infty$

Definition 3.15 — Divergence to $\pm\infty$

Let $\{s_n\}$ be a sequence of real numbers.

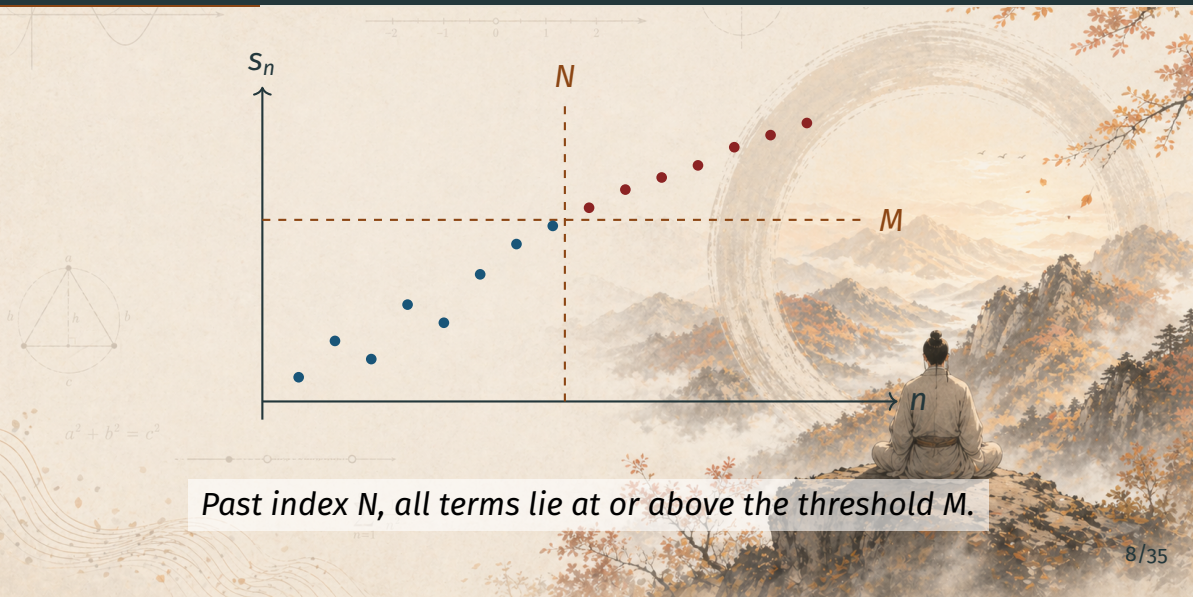
- $s_n \rightarrow +\infty$ if $\forall M \in \mathbb{R}, \exists N$ with

$$n \geq N \implies s_n \geq M.$$

- $s_n \rightarrow -\infty$ if $\forall M \in \mathbb{R}, \exists N$ with

$$n \geq N \implies s_n \leq M.$$

Picture: $s_n \rightarrow +\infty$



Past index N , all terms lie at or above the threshold M .

Remark on the Notation $\rightarrow$

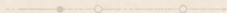
Remark

Definition 3.15 reuses the symbol $\rightarrow$ for certain **divergent** sequences, in addition to convergent ones.

The definition of convergence and limit (Def. 3.1) is *in no way changed*.



$$a^2 + b^2 = c^2$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

Upper and Lower Limits



Definition 3.16: The Set E of Subsequential Limits

Let $\{s_n\}$ be a real sequence. Define

The Set E

$$E = \{x \in [-\infty, +\infty] : s_{n_k} \rightarrow x \text{ for some subsequence } \{s_{n_k}\}\}.$$

E contains all real subsequential limits (Def. 3.5), **plus possibly** $\pm\infty$.

Definition 3.16: Upper and Lower Limits

Definition 3.16 — Upper and Lower Limits

$$s^* = \sup E, \quad s_* = \inf E.$$

The numbers s^* and s_* are the **upper** and **lower limits** of $\{s_n\}$.

Sup and inf are taken in $[-\infty, +\infty]$ (Definitions 1.8 and 1.23).

Definition 3.16: Notation $\limsup$ and $\liminf$

Definition 3.16 — Notation

$$\limsup_{n \rightarrow \infty} s_n = s^*, \quad \liminf_{n \rightarrow \infty} s_n = s_*.$$

Read as: the *limit superior* (or *upper limit*) and the *limit inferior* (or *lower limit*) of $\{s_n\}$.

Picture: E , $\limsup$, and $\liminf$

$$\liminf s_n = s_* = -2 \qquad \limsup s_n = s^* = 2$$



E (subsequential limits)

$\liminf$ is the smallest, $\limsup$ the largest subsequential limit.

Observation: E Is Always Non-Empty

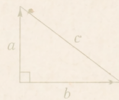
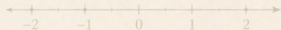
Observation

For every real sequence $\{s_n\}$, the set E is **non-empty**:
so $\limsup s_n$ and $\liminf s_n$ are always well-defined.

Three cases

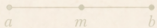
- (1) $\{s_n\}$ *unbounded above* $\implies$ some $s_{n_k} \rightarrow +\infty$. So $+\infty \in E$.
- (2) $\{s_n\}$ *unbounded below* $\implies$ some $s_{n_k} \rightarrow -\infty$. So $-\infty \in E$.
- (3) $\{s_n\}$ *bounded in $\mathbb{R}$* : all terms lie in a compact interval $[L, U]$. By Bolzano-Weierstrass Theorem 3.6, there exists a subsequence $s_{n_k} \rightarrow p \in [L, U]$. So $p \in E$. $\square$

$$f(x) = \int_a^x f(t) dt$$



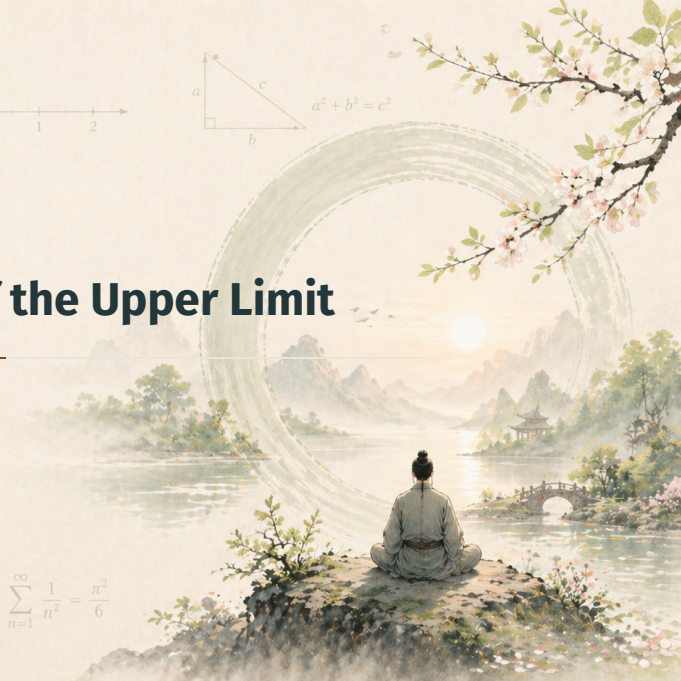
$$a^2 + b^2 = c^2$$

Properties of the Upper Limit



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

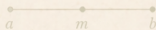
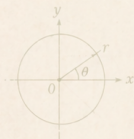


Theorem 3.17: Two Properties of s^*

Let $\{s_n\}$ be a real sequence; E and s^* as in Definition 3.16.

Theorem 3.17

(a) $s^* \in E$.



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

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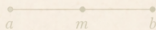
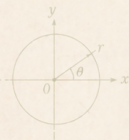
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Let $\{s_n\}$ be a real sequence; E and s^* as in Definition 3.16.

Theorem 3.17

- (a) $s^* \in E$.
- (b) If $x > s^*$, there is N such that $n \geq N$ implies $s_n < x$.



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$a^2 + b^2 = c^2$$

Theorem 3.17: Two Properties of s^*

Let $\{s_n\}$ be a real sequence; E and s^* as in Definition 3.16.

Theorem 3.17

(a) $s^* \in E$.

(b) If $x > s^*$, there is N such that $n \geq N$ implies $s_n < x$.

Moreover, s^* is the **only** number with properties (a) and (b).

An analogous statement holds for s_* .

Picture: The Two Properties of s^*



- (a) s^* is reached by some subsequence.
- (b) Nothing above s^* is reached infinitely often.

Proof Strategy: Theorem 3.17

Goal: (a) $s^* \in E$; (b) $s_n < x$ eventually for $x > s^*$; and uniqueness.

Plan for (a): *three cases on the value of s^* .*

- $s^* = +\infty$: $\{s_n\}$ unbounded above $\implies$ subsequence $\rightarrow +\infty$.
- s^* real: invoke Theorems 3.7 and 2.28.
- $s^* = -\infty$: only $-\infty$ in E ; $s_n \rightarrow -\infty$.

Plan for (b): $s_n \geq x$ infinitely often forces a $y \in E$ with $y \geq x > s^*$, which contradicts $s^* = \sup E$.

Uniqueness: if $p < q$ both work, an $x \in (p, q)$ contradicts (a) at q .

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Proof of (a): $s^* \in E$ [Case $s^* = +\infty$]

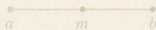
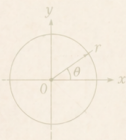
$$s^* = +\infty$$

$s^* = \sup E = +\infty$ implies that $\{s_n\}$ is unbounded above.

(If $\{s_n\}$ was bounded above, E would also be bounded above.)

Thus there exists a subsequence $\{s_{n_k}\}$ such that $s_{n_k} \rightarrow +\infty$.

Therefore $+\infty \in E$, that is, $s^* \in E$.



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$a^2 + b^2 = c^2$$

Proof of (a): $s^* \in E$ [Case s^* Real]

s^* real

E is bounded above and at least one real subsequential limit exists.

- Theorem 3.7: real subsequential limits form a closed subset of $\mathbb{R}$.
- Theorem 2.28: sup of a closed, bounded-above subset of $\mathbb{R}$ lies in that subset.

Hence $s^* = \sup E \in E$.

Proof of (a): $s^* \in E$ [Case $s^* = -\infty$]

$$s^* = -\infty$$

$\sup E = -\infty$ forces $E = \{-\infty\}$.

Hence $+\infty \notin E$, so $\{s_n\}$ is bounded above; and no *real* subsequential limit exists.

Fix any real M . If $s_n > M$ for infinitely many n , then that subsequence is bounded (both above by the bound on $\{s_n\}$, and below by M), so by Bolzano-Weierstrass (Theorem 3.6) it has a real subsequential limit $y \geq M$, **contradicting** $E = \{-\infty\}$.

Thus $s_n \leq M$ eventually, that is, $s_n \rightarrow -\infty$. Hence $s^* = -\infty \in E$. $\square$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

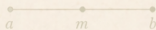
Proof of (b): $s_n < x$ eventually for $x > s^*$

If $x > s^*$, then $s_n < x$ eventually

Suppose, for contradiction, $s_n \geq x$ for infinitely many n . Restrict to that subsequence; all its terms are $\geq x$.

- If *bounded above*: Bolzano-Weierstrass (Theorem 3.6) gives a sub-subsequence with real limit $y \geq x$.
- If *unbounded above*: a sub-subsequence diverges to $y = +\infty \geq x$.

Either way, $y \in E$ with $y \geq x > s^* = \sup E$, **contradiction**. $\square$


$$a \quad m \quad b$$


$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$a^2 + b^2 = c^2$$

Proof of Uniqueness

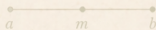
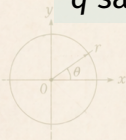
Only s^* satisfies (a) and (b)

Suppose p and q both satisfy (a) and (b), with $p < q$.

Choose x with $p < x < q$.

p satisfies (b): $s_n < x$ for all $n \geq N$.

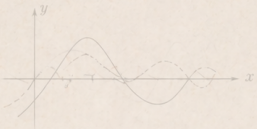
q satisfies (a): some subsequence converges to $q > x$. **Impossible.** $\square$



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$a^2 + b^2 = c^2$$

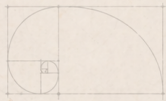


$$e^{i\theta} = \cos \theta + i \sin \theta$$



$$\nabla^2 \phi = 0$$

Examples



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$



Example 3.18 (a): A Sequence Hitting Every Rational

Example 3.18 (a)

Let $\{s_n\}$ enumerate **all rational numbers**.

Every real number is a subsequential limit, so $E = [-\infty, +\infty]$.

$$\limsup_{n \rightarrow \infty} s_n = +\infty, \quad \liminf_{n \rightarrow \infty} s_n = -\infty.$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Example 3.18 (b): An Oscillating Sequence

Example 3.18 (b)

Let

$$s_n = \frac{(-1)^n}{1 + 1/n}.$$

Then

$$\limsup_{n \rightarrow \infty} s_n = 1, \quad \liminf_{n \rightarrow \infty} s_n = -1.$$

Why: the even- n subsequence $\rightarrow +1$; the odd- n subsequence $\rightarrow -1$.

Picture: Example 3.18 (b)

S_n



n



2

4

6

8

10

$+1$

-1

$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$

$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$

Two interlaced subsequences with limits ± 1 .

Example 3.18 (c): Convergence Criterion

Example 3.18 (c)

For a real sequence $\{s_n\}$:

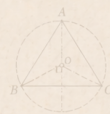
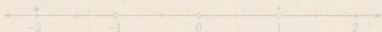
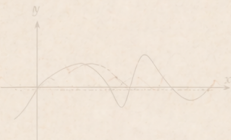
$$\lim_{n \rightarrow \infty} s_n = s \iff \limsup_{n \rightarrow \infty} s_n = \liminf_{n \rightarrow \infty} s_n = s.$$

Why this matters

The ordinary limit exists **exactly when** $\limsup$ and $\liminf$ **agree**.
All asymptotic information is captured by the two extremes.

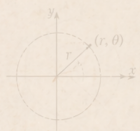
$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$



$$a^2 + b^2 = c^2$$

Comparing Sequences



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Theorem 3.19: Order Is Preserved

Theorem 3.19

If $s_n \leq t_n$ for $n \geq N$ (some fixed N), then

$$\liminf_{n \rightarrow \infty} s_n \leq \liminf_{n \rightarrow \infty} t_n,$$

$$\limsup_{n \rightarrow \infty} s_n \leq \limsup_{n \rightarrow \infty} t_n.$$

Pointwise inequality of tails $\implies$ inequality of upper/lower limits.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Proof of Theorem 3.19 (limsup case)

Let $s^* = \limsup_{n \rightarrow \infty} s_n$, $t^* = \limsup_{n \rightarrow \infty} t_n$. Goal: $s^* \leq t^*$.

Proof

Case $t^* = +\infty$: Trivially $s^* \leq +\infty = t^*$.

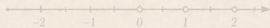
Otherwise, fix any real $x > t^*$. By Theorem 3.17 (b) applied to $\{t_n\}$:

$$\exists N_1 : n \geq N_1 \implies t_n < x.$$

Combined with $s_n \leq t_n$ for $n \geq N$: $s_n \leq t_n < x$ for all $n \geq \max(N, N_1)$.

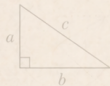
Hence every subsequential limit of $\{s_n\}$ is $\leq x$, so $s^* \leq x$.

Since $x > t^*$ was arbitrary: $s^* \leq t^*$. $\square$

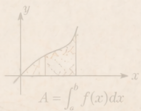


$$x^2 + y^2 = r^2$$

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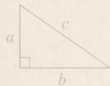
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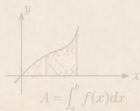
Summary

Key takeaways from this section:

- **Def. 3.15:** $s_n \rightarrow \pm\infty$ defined for divergent sequences



$$a^2 + b^2 = c^2$$

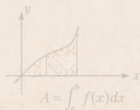
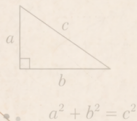


$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

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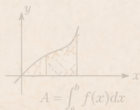
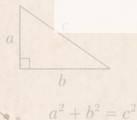


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- **Thm. 3.17:** s^* is the *unique* number that lies in E and beyond which the tail does not return



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

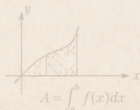
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- **Thm. 3.17:** s^* is the *unique* number that lies in E and beyond which the tail does not return
- **Convergence criterion:** $\lim s_n = s \iff \limsup s_n = \liminf s_n = s$
- **Thm. 3.19:** $s_n \leq t_n$ eventually $\implies$ inequalities for $\limsup$ and $\liminf$



$$a^2 + b^2 = c^2$$

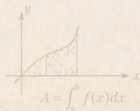
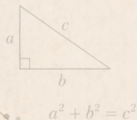


$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Coming Up Next

Next section: Some Special Sequences

- Concrete sequences whose limits we can compute
- n^p , p^n , $n^{1/n}$, and friends
- Tools we will reuse for series and analysis



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$